

HOUSE PRICE PREDICTION USING RANDOM FOREST IN SOUTH JAKARTA

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Abstrak

This study aims to predict house prices in Tebet, South Jakarta by using multiple linear regression based on factors such as building area, land area, number of bedrooms and bathrooms, and garage facilities and using the Random Forest method to improve the accuracy and relevance of the model in the hope of providing guidance for buyers and property agents in dealing with prices in increasingly tight market fluctuations.

Keywords : fluctuation, house, Jakarta, linear, price, regression

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1. INTRODUCTION

Housing or a home is one of the fundamental human needs that serves as a place to live, relax, gather with family, shelter, and rest after daily activities[1][2]. This makes people consider whether the house they purchase will provide a good return on investment, as home prices continue to rise over time. The better the quality of the house and the more strategic the location, the higher the likelihood that the house's price will be higher[3][4][5].

The price of houses continues to rise every year, and this increase can be measured through various aspects or factors that affect the price offered[5]. Prices are not always certain and accurately predicted, which makes home buyers need a system that can predict house prices based on the supporting factors[6]. In this context, the utilization of machine learning methods offers great potential to provide more accurate predictions compared to conventional methods (Friedman et al., 2001). In relation to this, research was conducted using a regression algorithm to calculate the house price specifications using Multiple Linear Regression and analyzing data to build a machine learning model aimed at predicting house prices[7][8].

Machine Learning is a part of Artificial Intelligence (AI) that is used to develop systems capable of learning independently without the need for repeated programming[9]. Machine Learning can reduce the burden on humans to explain and formulate their knowledge into a form understandable by machines, as well as enable the development of intelligent systems more efficiently[10]. The system to be used in building the machine learning model in this research is an Artificial Neural Network. Artificial Neural Networks are computational models inspired by the biological neural networks found in humans. The advantage of this model lies in its ability to learn complex patterns that are difficult to identify using traditional methods[11]. Artificial Neural Networks are developed to solve problems by understanding learning process patterns or identifying classifications[12][13].

The Multiple Linear Regression (MLR) and Random Forest methods are two approaches commonly used in house price prediction. MLR is a simple yet effective statistical technique for modeling the linear relationship between independent and dependent variables. In the context of house price prediction, MLR can be used to identify the influence of factors such as land area, number of rooms, and location on the house price. The main advantage of MLR is its simplicity and the ease of interpreting the model's results[14].

Linear regression as a prediction method has been used to predict book sales and inventory goods[15] for future periods. Research conducted by Wahyuni, Nafi'iyah, and Masruroh discussed a housing sales forecasting system using multiple linear regression with an MSE value of 5.557[16]. Housing issues are significant in many countries, including Indonesia. Factors such as population growth, urbanization, and land scarcity have driven up housing prices, especially in urban areas. Therefore, the ability to predict house prices with high accuracy becomes a critical need for buyers, sellers, and property developers[17].

Random Forest is an ensemble algorithm based on decision trees that combines predictions from multiple trees to improve accuracy and reduce overfitting[18]. This algorithm is effective in capturing non-linear relationships, interactions between variables, and handling datasets with many features and missing data. Random Forest is often used for predictions, such as house prices, due to its ability to handle complex and dynamic data[12][13][14].

This research aims to evaluate the performance of MLR and Random Forest in predicting simple house prices. The dataset used includes key variables such as land area, number of rooms, number of bathrooms, garage area and number of floors[5][15]. This approach allows for in-depth analysis of the advantages and limitations of each method, providing recommendations for suitable models to develop property price prediction systems (Darmawan et al., 2020; Zhang et al., 2021). The results of this research are expected to contribute to the development of more accurate and efficient house price prediction systems. By utilizing machine learning technologies like MLR and Random Forest, it is hoped that the general public and property industry players can use these predictions for better decision-making[23][24]. Additionally, this research also provides a foundation for developing technology-based applications for direct house price estimation, potentially supporting transaction efficiency in the property market[25].

2. METHOD

The method employed in this study uses a quantitative approach through Multiple Linear Regression and Random Forest by analyzing house price data in the Tebet area, South Jakarta. The primary goal of this research is to develop a predictive model capable of estimating current and future house prices based on relevant variables such as building area, land area, and other influencing factors.

With a 5% error level ($\alpha = 0.05$), the results have a 95% chance of representing the actual relationship in the population, making it reliable for practical conclusions. This level is also more efficient compared to 99% because it requires a smaller sample size, saving time and resources. In addition, many statistical software use 95% as the default setting, making interpretation easier and increasing consistency with existing literature.

2.1 Data

The data used in this study was sourced from Kaggle, verified for authenticity. The dataset includes detailed information about houses, covering the following variables:

1. Building area
2. Land area
3. Number of bedrooms

4. Number of bathrooms
5. Garage size
6. Number of floors
7. Historical house price increases from Bank Indonesia

The variables in this study consist of independent variables (e.g., building area, land area, etc.) and a dependent variable, which is the house price itself[5][26].

2.2 Research Stages

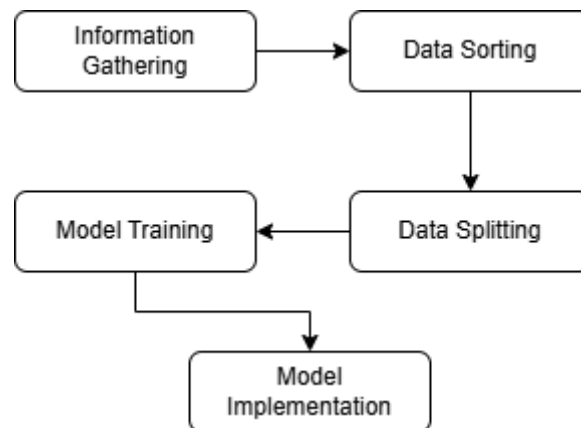


Figure 1. Research Diagram.

Information gathering was conducted by searching through secondary data available on the internet. The data found was then sorted to ensure only relevant and required data was used. During the data splitting stage, 80% of the data obtained was allocated as training data, while the remaining 20% was used as testing data. The finalized model was trained using the separated data, and the resulting model was implemented into a simple Python application.

2.3 Random Forest

The Random Forest algorithm is an extension of the Classification and Regression Tree (CART) method, which applies bootstrap aggregating (bagging) and random feature selection techniques [17]. Random Forest is a method used for classification tasks by constructing multiple classification trees. This approach enhances accuracy by generating child nodes for each parent node through random selection.

The classification results from each tree are aggregated, and the most frequent classification outcome is chosen. The method consists of root nodes, internal nodes, and leaf nodes:

- **Root Node:** The topmost node, serving as the starting point of the decision tree.
- **Internal Node:** A branching node with at least one input and two or more outputs.
- **Leaf Node:** The terminal node, which has one input and no output.

The decision tree process begins by calculating the entropy value to determine the impurity level of an attribute and the information gain value. Entropy is calculated using the formula in Equation (1), while information gain is determined using Equation (2) [25].

Entropy

$$\text{Entropy}(Y) = - \sum p(c|Y) \log_2 p(c|Y) \quad \text{Entropy}(Y) = - \sum p(c|Y) \log_2 p(c|Y)$$

Description:

- YY: Set of cases
- $p(c|Y)$: Proportion of YY values in class cc

Information Gain

$$\text{Information Gain}(Y, a) = \text{Entropy}(Y) - \sum_{v \in \text{Values}(a)} \frac{|Y_v|}{|Y|} \text{Entropy}(Y_v)$$

$$\text{Gain}(Y, a) = \text{Entropy}(Y) - \sum_{v \in \text{Values}(a)} \frac{|Y_v|}{|Y|} \text{Entropy}(Y_v)$$

Description:

- Values(a): Possible values in the set of cases for attribute aa
- Y_v : Subset of YY with class YY related to aa
- Y_a : All values of YY corresponding to aa

2.4 Multiple Linear Regression

Multiple Linear Regression is a regression model with more than one variable, where the multiple linear regression modeling predicts using interval or ratio scale data and involves more than one predictor. Additionally, this method analyzes the relationship between independent variables and a dependent variable [17].

The multiple linear regression model can be represented by the following equation:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + e$$

Description:

- YY: Dependent variable
- XX: Independent variable
- α : Constant (Intercept)
- β : Slope or coefficient estimate
- ee: Error

3 RESULT**3.1 Data Preprocessing**

The research dataset includes **1,010 house records** with 7 key variables, namely:

HARGA	LB	LT	KT	KM	GRS	LNT
3800000000	220	220	3	3	0	1
4600000000	180	137	4	3	2	2
3000000000	267	250	4	4	4	2
4300000000	40	25	2	2	0	2
9000000000	400	355	6	5	3	2

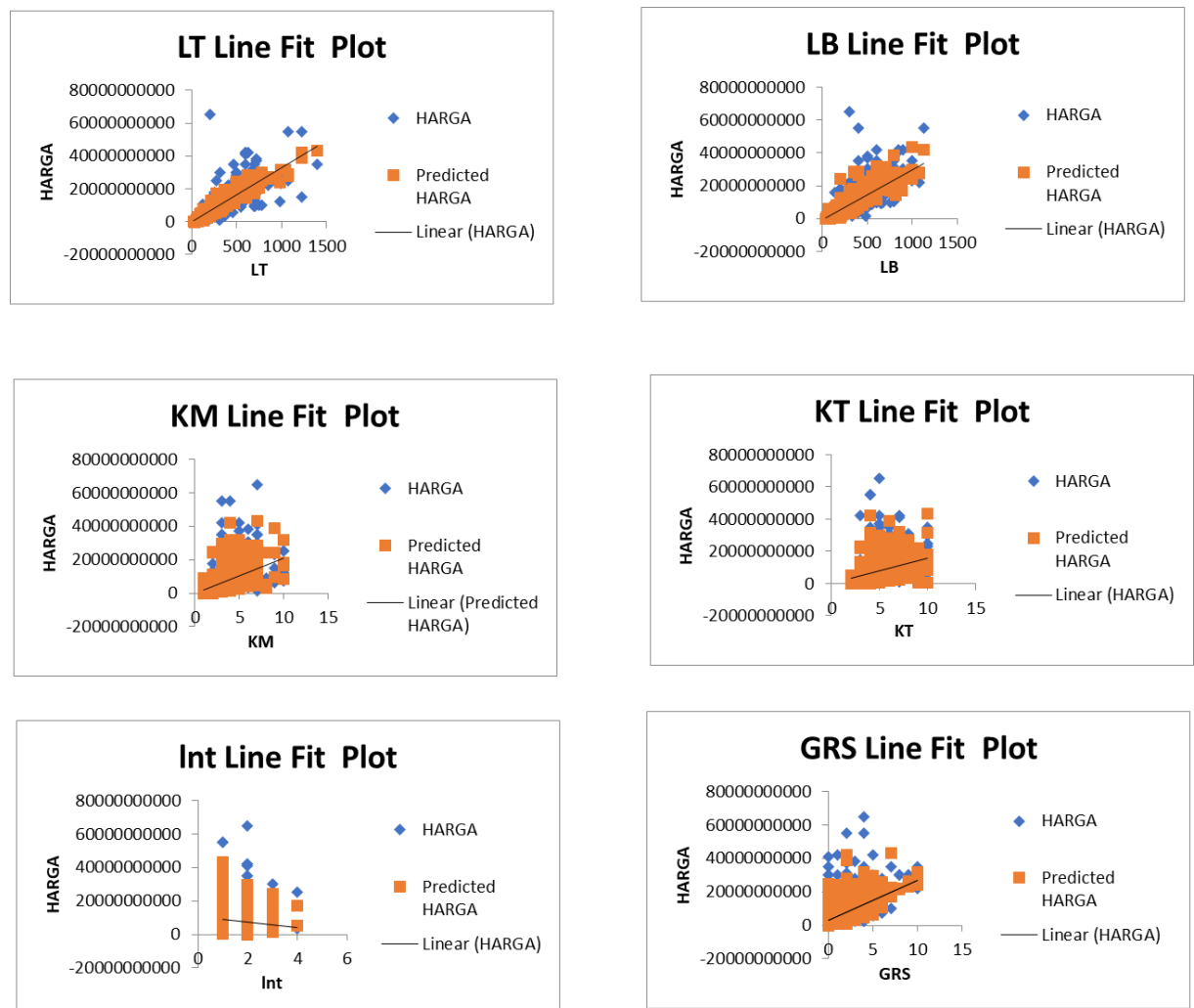
Figure 2. Cleaned Data.

1. **Price:** House price (dependent variable).
2. **LB:** Building area (m²).
3. **LT:** Land area (m²).
4. **KT:** Number of bedrooms.
5. **KM:** Number of bathrooms.
6. **GRS:** Garage size (m²).
7. **LNT:** Number of floors.

The data has been cleaned and is free of missing values, making it ready for further analysis.

3.2 Data Analysis

The study utilized a **Line Fit Plot** to visualize the relationship between the dependent variable (*target*) and independent variables (*predictors*).

**Figure 3.** Line Fit Plot.

Visualization Results:

- The variables **land area (LT)** and **building area (LB)** showed the strongest and most consistent influence on house prices.
- Other variables, such as **number of bathrooms (KM)**, **bedrooms (KT)**, **garage size (GRS)**, and **number of floors (LNT)**, also influenced prices but with higher variation.

3.3 Data Splitting

- **Training Data:** 80% (808 records) used to train the model.
- **Testing Data:** 20% (202 records) used to test the model's accuracy.

3.4 Model Statistics

3.4.1 Regression Statistics

The model evaluation metrics are shown in the table below:

Regression Statistics	
R Square	0,82
Root Min Square Error	0,31
Min Absolute Error	0,22
Observations	1010

Table 1. Regression Statistics

Interpretation:

- The **R Square (0.82)** value indicates that the model explains 82% of the variation in house prices based on the independent variables.
- The **RMSE (0.31)** and **MAE (0.22)** values indicate a small prediction error, reflecting the model's consistency with the actual data.

3.4.2 Analysis of Variance (ANOVA)

The ANOVA results are as follows:

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	6	3,8993E+22	6,5E+21	423,7775	6,2087E-271
Residual	1003	1,53815E+22	1,53E+19		
Total	1009	5,43745E+22			

Table 2. ANOVA

Conclusion: The model is statistically significant, with an **F-statistic (423.7775)** and a very small **Significance F** value.

3.4.3 Regression Coefficients

The regression coefficient values for each variable are shown in the table below:

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95,0%	Upper 95,0%
Intercept	134705925,4	636181144	0,2117	0,832352	-1113692671	1383104522	-1113692671	1383104522
LB	13978336,14	1473759,074	9,4848	1,7E-20	11086331,6	16870340,68	11086331,6	16870340,68
LT	21740427,61	1518792,137	14,314	1,99E-42	18760053,25	24720801,98	18760053,25	24720801,98
KT	-626148109	108905555,3	-5,7494	1,19E-08	-839856961,7	-412439256,2	-839856961,7	-412439256,2
KM	481409415,9	127317414,4	3,7811	0,000165	231570383,5	731248448,3	231570383,5	731248448,3
GRS	325129247,7	96858055	3,3567	0,000818	135061590	515196905,4	135061590	515196905,4
Int	-552178004,6	312032612,7	-1,7696	0,077095	-1164489576	60133566,48	-1164489576	60133566,48

Table 3. Regression Coefisien

The variables **LB**, **LT**, **KM**, and **GRS** have a significant positive contribution to house prices.

- The variable **KT** shows a significant negative contribution.
- The variable **LNT** is not significant at a 95% confidence level.

3.5 Python Graphical User Interface (GUI)

The Graphical User Interface (GUI) is built using the Tkinter library to facilitate user interaction with the house price prediction system. The interface consists of various elements such as labels, entry fields, and buttons, which make it easy for users to input property data (e.g., building area, number of rooms, etc.) and obtain predicted house prices for the years 2020, 2024, and other specific years.

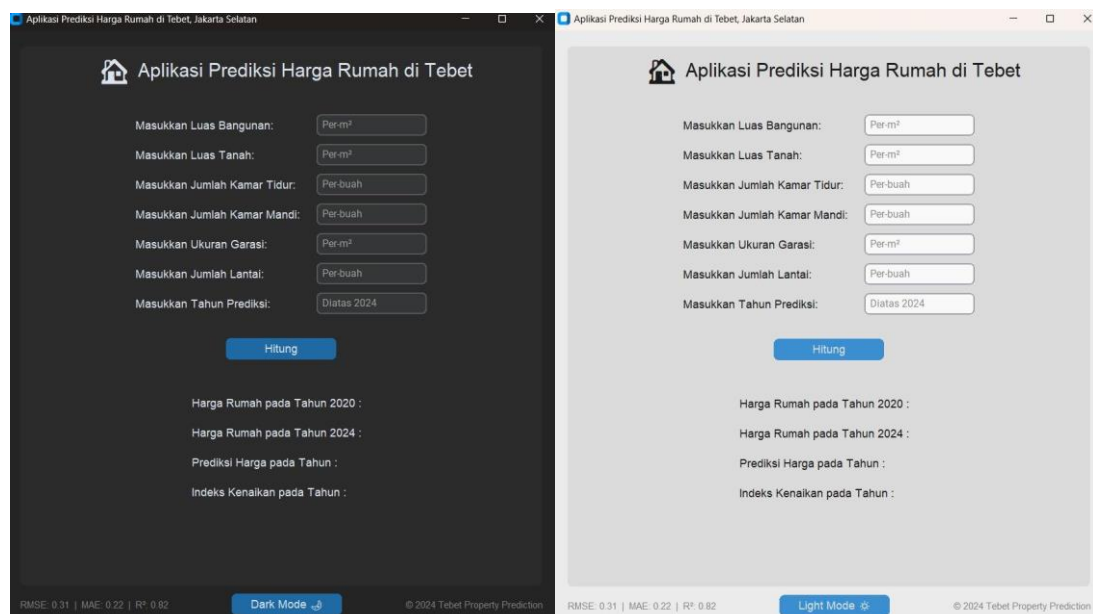


Figure 4. Pre-input Interface.

In this application, users can enter house criteria in the form displayed on the application screen. After entering the house criteria to be predicted, users can press the "Calculate" button to run the application and produce an output for the house price based on the input criteria. The output display can be seen in **Figure 5**.

Figure 5. Output Interface

For users who want to make a new prediction, they can simply modify the input criteria in the form.

4 DISCUSSIONS

This study shows that Multiple Linear Regression (MLR) is effective in predicting house prices in Tebet, with land area (LT) and building area (LB) as the most significant factors, achieving high accuracy with an R Square of 0.82. However, data from Kaggle does not fully reflect local market conditions, suggesting that combining additional methods such as Random Forest could improve prediction results. The Analysis of Variance (ANOVA) results show a Significance F of less than 5%, a common indicator in linear regression, confirming that these variables significantly affect house prices in Tebet, South Jakarta. Land area emerges as the most dominant factor (p-value far below 5%), while the number of floors (LNT) has the smallest effect (p-value > 5%).

To improve the quality of the prediction model, a more comprehensive combination of variables from the two previous studies is proposed. These variables include structural aspects (LT, LB, number of bedrooms, bathrooms, garage capacity, house size, property type), as well as location and environmental factors (location, distance to shopping centers, schools, hospitals, access to public transportation). Additional variables such as furniture status (furnishing) and accessibility to extra facilities are also included to broaden the model's perspective.

In building the model, an ensemble approach is used by combining XGBoost, Random Forest Regression, LightGBM, and MLR as the baseline model. Techniques such as stacking, blending, and regressor voting are applied to enhance prediction accuracy. The model evaluation is conducted using an 80:20 train-test split, cross-validation, and performance metrics including Mean Absolute Error

(MAE), Root Mean Squared Error (RMSE), and R-squared. The model's performance is further optimized using GridSearchCV or RandomizedSearchCV.

This integrated model not only improves the accuracy of house price predictions but also provides valuable insights for consumers and developers in making more informed pricing decisions in the real estate sector. By leveraging advanced machine learning techniques and a broad set of predictive variables, the model offers a robust tool for navigating the complexities of the housing market.

5 CONCLUSION

This study successfully developed a house price prediction model in Tebet, South Jakarta, using the Multiple Linear Regression (MLR) method. The research results show that land area and building size are the most significant factors in determining house prices, with a high accuracy rate (R-squared 0.82). However, the data used from Kaggle has limitations because it does not specifically reflect the local market. This study provides important contributions in helping consumers and property developers make decisions based on objective price estimates. For further development, the use of primary data and exploration of other methods, such as Random Forest, could enhance the accuracy and relevance of the model.

The number of rooms in a house can negatively affect the price for several reasons. First, houses with too many rooms are often considered inefficient in space utilization, reducing their appeal to buyers. Additionally, houses with many rooms might be located in less strategic areas or have smaller total sizes, which can lower the overall value.

CONFLICT OF INTEREST

The author declares that there is no conflict of interest between the authors and the subject of the research in the writing of this paper.

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